



Project Proposal

Arunella

CSC 313 1.5 Service Oriented Computing

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1. Introduction

Agriculture plays a vital role in Sri Lanka's economy by supporting livelihoods and contributing to national food security. Despite this importance, the agricultural sector continues to face a range of systemic challenges that limit its full potential.

Farmers typically rely on traditional selling methods and intermediaries (middlemen) to distribute their produce. This dependency restricts their access to fair market pricing and limits their reach to wider consumer markets. On the buyer side, sourcing reliable suppliers and obtaining fresh produce at competitive prices remains difficult due to the lack of a transparent, connected marketplace.

Coordination between the three key stakeholder groups: farmers, buyers, and transporters, is often fragmented, leading to transportation delays, inefficient order management, and significant wastage of perishable agricultural products.

This project proposes a Smart Agricultural Supply Chain Management System that digitally connects all stakeholders through a centralized platform. The system is delivered as two distinct interfaces: a Mobile Application serving farmers, buyers, and transporters, and an Admin Website serving platform administrators, improving transparency, coordination, and overall efficiency in agricultural product distribution across Sri Lanka

2. Problem Statement

Despite the critical role agriculture plays in Sri Lanka's national economy, the current agricultural supply chain remains heavily reliant on traditional and manual processes. This has created a range of interconnected challenges that negatively impact all stakeholders in the value chain.

2.1 Challenges for Farmers

- Heavy dependence on intermediaries and local markets restricts access to fair pricing and broader market opportunities.
- Reduced income and financial instability resulting from limited bargaining power against middlemen.
- No reliable mechanism to advertise available crops, quantities, or pricing to a wider audience.

2.2 Challenges for Buyers

- Difficulty in locating reliable suppliers and obtaining fresh produce at reasonable prices.
- Lack of a transparent, connected marketplace makes price comparison and supplier evaluation inefficient.
- No real-time visibility into product availability or order fulfilment status.

2.3 Operational & Systemic Challenges

- Poor coordination between farmers, buyers, and transporters causes transportation delays.
- Inefficient order management leads to wastage of perishable agricultural products.
- Absence of delivery status updates.
- Limited monitoring capabilities for administrators, increasing the risk of fraud and supply-demand mismatches.
- Increased miscommunication and operational inefficiencies due to the lack of a centralized system.

These challenges collectively underline the urgent need for a centralized digital platform that connects all stakeholders and efficiently manages agricultural product distribution through improved communication, coordination, and monitoring.

3. Project Objectives

Arunella is designed to achieve the following strategic objectives:

Objective 1: Manage Farmer and Crop Information

Create a comprehensive system for registering farmers and maintaining detailed crop records. Farmers will be able to add, update, and manage information about their harvests, including crop type, available quantity, price per kilogram, expiry date, and geographic location. This provides a reliable, searchable inventory of available agricultural produce.

Objective 2: Enable Market Access and Order Management

Empower buyers to view available crops and place orders directly eliminating dependency on intermediaries. This ensures fair pricing, transparent transactions, and a more efficient connection between supply and demand in the agricultural market.

Objective 3: Improve Logistics and Delivery Management

Efficiently manage the entire transportation and delivery lifecycle. Transporters will be assigned delivery tasks and update delivery status at each stage. This ensures timely distribution, reduces transportation delays, and minimizes losses from mishandled or delayed deliveries.

Objective 4: Provide Secure Administration and Monitoring

Equip administrators with comprehensive tools to manage users, monitor products and deliveries, prevent fraudulent activities, and proactively handle supply-demand imbalances. The Admin Service provides a centralized oversight layer to maintain system integrity and operational health.

4. Scope of the System

Arunella will be developed as two dedicated platform interfaces, each tailored to its audience:

Service Module	Key Capabilities
Mobile App <i>(Farmers, Buyers & Transporters)</i>	Farmer registration and crop listing management Buyer marketplace for browsing, and ordering crops Transporter task assignment Track order status and delivery updates
Admin Website <i>(Administrators only)</i>	Centralized user, product, and delivery monitoring Fraud detection and supply-demand management tools System-wide oversight and configuration management

Table 1: Scope of the system

5. Database Design

The system uses a relational database architecture to manage all entities and their relationships. The schema is designed to ensure data integrity, scalability, and efficient querying across all platform services.

5.1 Entity Overview

The database comprises eight core entities:

Entity	Primary Key	Description
ADMIN	admin_id	Platform administrators with management privileges
FARMER	user_id	Registered farmers with crop listing capabilities
BUYER	user_id	Buyers who browse listings and place orders

TRANSPORTER	user_id	Transporters assigned delivery tasks
CROP	crop_id	Crop listings with pricing, stock, and metadata
ORDER	order_id	Purchase orders placed by buyers
DELIVERY	delivery_id	Delivery records linked to orders and transporters
HAS	order_id, crop_id	Junction table linking orders to specific crops

Table 2: Primary key table

5.2 Key Attributes by Entity

ADMIN: (admin_id [PK], name, email, password)

FARMER: (user_id [PK], admin_id [FK], role, name, email, password, nic, contact_no, district, rating, location, wallet, bankA/C_no)

BUYER: (user_id [PK], admin_id [FK], name, email, password, nic, contact_no, district, rating, role, businessRegNo, market_location)

TRANSPORTER: (user_id [PK], admin_id [FK], name, email, password, nic, contact_no, district, rating, role, vehiclePlateNo, max_capacity)

CROP: (product_id [PK], product_name, user_id [FK], price_per_kg, stock, status, uploaded_date, exp_date, min_price, description, image)

ORDER: (order_id [PK], user_id [FK], price, quantity, date, status)

DELIVERY: (delivery_id [PK], user_id [FK], order_id [FK], pickup_location, delivery_location, status, confirmation_img, date)

HAS: (order_id [PK][FK], product_id [PK][FK]) — composite primary key

6. EER Diagram

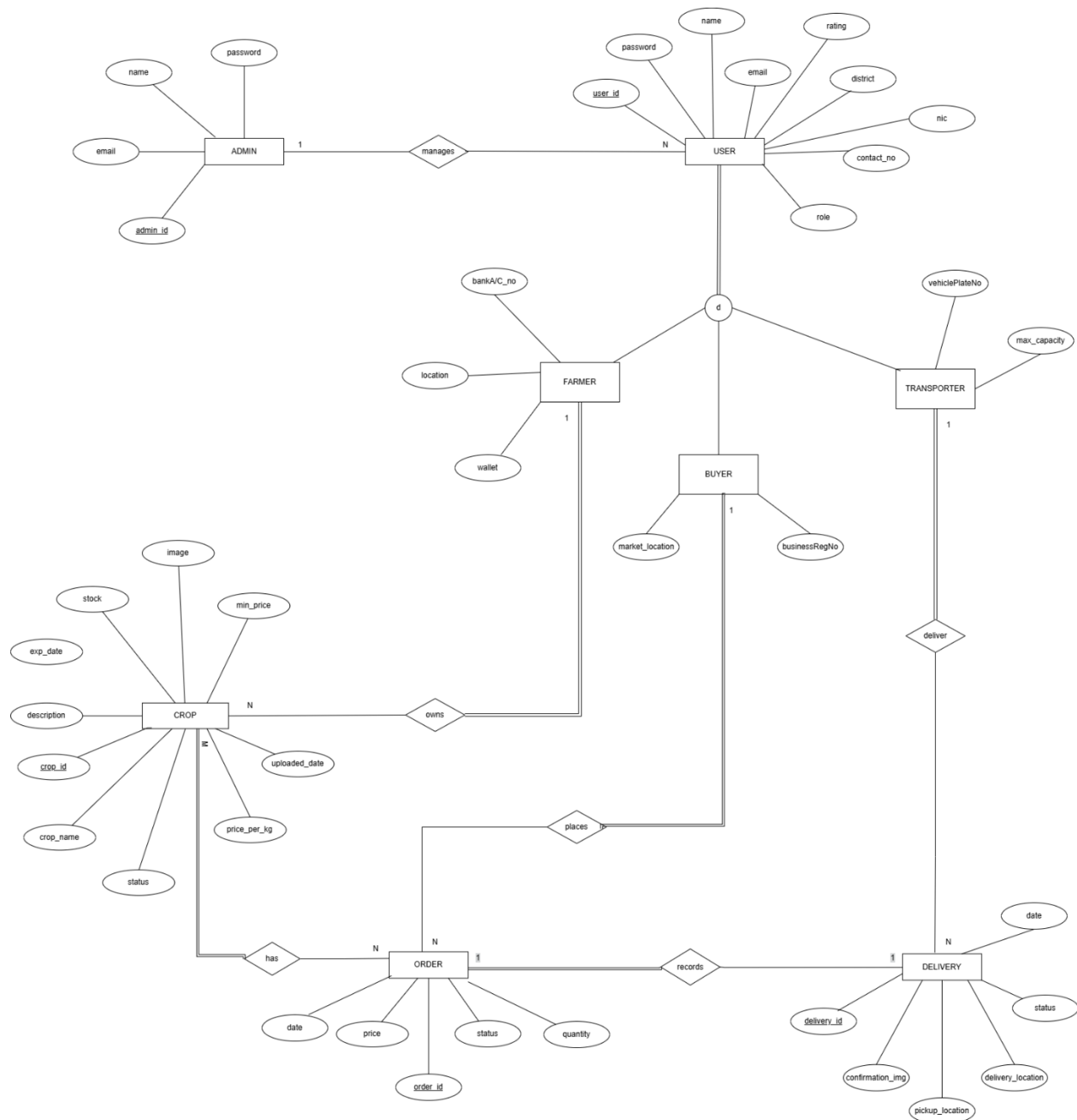


Figure 1: The ER conceptual schema diagram for the *Arunella* database

7. Schema Diagram

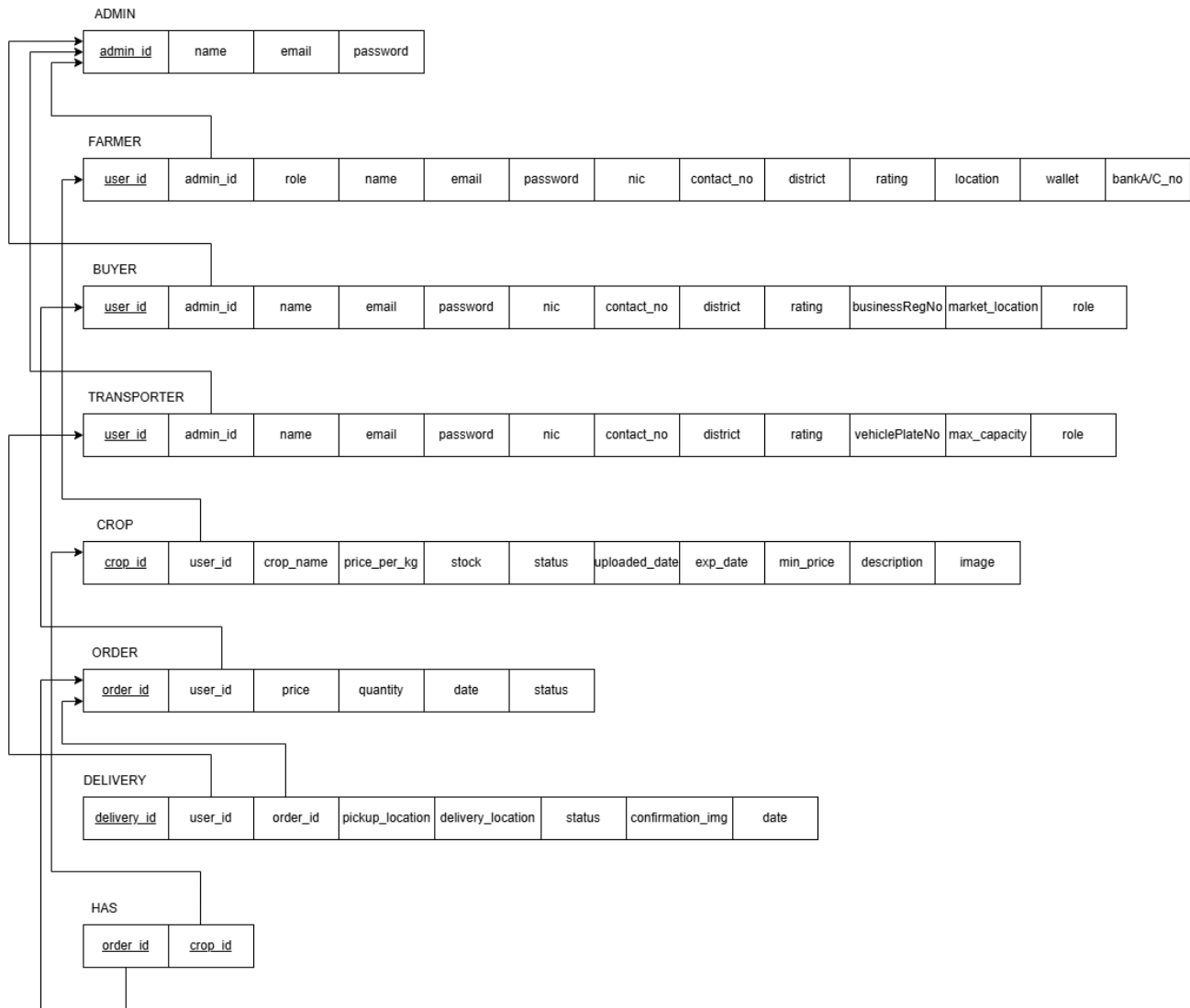


Figure 2: Result of mapping the *Arunella* ER schema into a relational schema

8. Conclusion

Arunella represents a timely and necessary intervention in Sri Lanka's agricultural sector. By replacing fragmented, manual supply chain processes with an integrated digital platform, the system will empower farmers, buyers, and transporters to collaborate more effectively, transparently, and profitably.

The proposed system directly addresses the key pain points of the current agricultural supply chain: poor market access, price opacity, logistics inefficiencies, and wastage through a carefully designed, scalable technology solution grounded in user-centered research.

With a clear scope, well-defined database architecture, and a structured development methodology, *Arunella* is poised to make a meaningful contribution to the modernization of agricultural commerce in Sri Lanka and serve as a model for similar initiatives across the region.